5	(a)	Increase in internal energy = work done on the system + heat supplied to the system. One mark for correct definition [M1] One mark for putting into a "word equation" [A1]			
	(b)	(i)	$u_C = u_A + \Delta u$ $= 45 + (-40 + 90)$ $= 95$ kJ		[C1] [A1]
		(ii)	$\Delta u = Q + W$ $-40 + 90 = Q - 20$ $Q = 70$ kJ		[C1] [A1]
		(iii)	1.	net work done on gas = -40 + 31 = -9.0 kJ	[C1] [A1]
		(iii)	2	Since there is no change in internal energy in a cycle, $\Delta u = 0$ $Q_{net} + W_{net} = 0$ $Q_{net} - 9.0 = 0$ $Q_{net} = 9.0$ kJ Ecf allowed.	[A1] [A1]